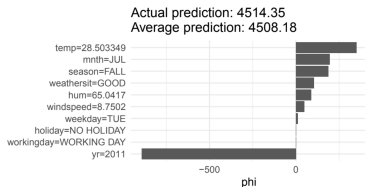
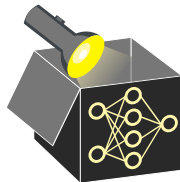


Interpretable Machine Learning

Shapley

Shapley Values for Local Explanations

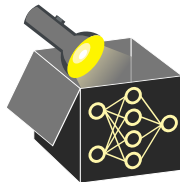
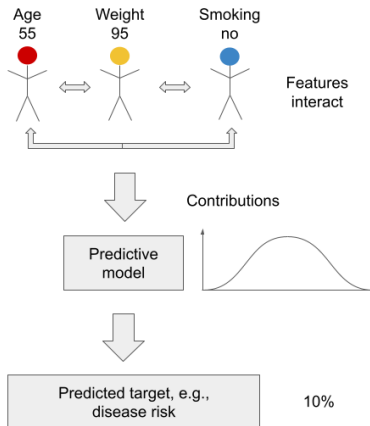


Learning goals

- See model predictions as a cooperative game
- Transfer the Shapley value concept from game theory to machine learning

FROM GAME THEORY TO MACHINE LEARNING

- Model prediction depends on feature interactions for a specific observation
- **Goal:** Decompose prediction into **individual feature contributions**
- **Idea:** Treat features as players jointly producing a prediction
- How to fairly assign credit to features?
⇒ Shapley values

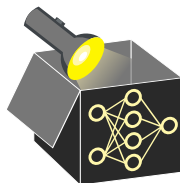


FROM GAME THEORY TO MACHINE LEARNING

- **Game:** Predict $\hat{f}(x_1, x_2, \dots, x_p)$ for a single observation $\mathbf{x}$
- **Players:** Features $x_j, j \in \{1, \dots, p\}$, cooperate to produce a prediction
- **Value function:** Defines payout of coalition $S \subseteq P$ for observation $\mathbf{x}$ by

$$v(S) = \hat{f}_S(\mathbf{x}_S) - \hat{f}_\emptyset, \text{ where}$$

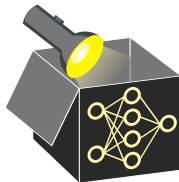
- $\hat{f}_S : \mathcal{X}_S \mapsto \mathcal{Y}$ is the PD function $\hat{f}_S(\mathbf{x}_S) := \int \hat{f}(\mathbf{x}_S, X_{-S}) d\mathbb{P}_{X_{-S}}$
 $\rightsquigarrow$ "Removes" features in $-S$ by marginalizing, keeping $\hat{f}$ fixed
- Mean prediction $\hat{f}_\emptyset := \mathbb{E}_{\mathbf{x}}(\hat{f}(\mathbf{x}))$ is subtracted to ensure $v(\emptyset) = 0$
- **Goal:** Distribute total payout $v(P) = \hat{f}(\mathbf{x}) - \hat{f}_\emptyset$ fairly among features



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- **Goal:** Distribute total payout $v(P) = \hat{f}(\mathbf{x}) - \hat{f}_\emptyset$ fairly among features
- **Marginal contribution of feature j joining coalition S** ($\hat{f}_\emptyset$ cancels):

$$\Delta(j, S) = v(S \cup \{j\}) - v(S) = \hat{f}_{S \cup \{j\}}(\mathbf{x}_{S \cup \{j\}}) - \hat{f}_S(\mathbf{x}_S)$$

- **Example (3 features):** Feature contributions for joining order $x_1 \rightarrow x_2 \rightarrow x_3$ toward total payout $v(P) = \hat{f}(\mathbf{x}) - \hat{f}_\emptyset$, each step reflects a marginal contribution



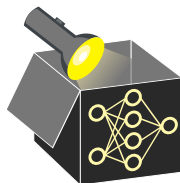
SHAPLEY VALUE - DEFINITION

► “Shapley” 1953

► “Strumbelj et al.” 2014

Order definition: Shapley value $\phi_j(\mathbf{x})$ quantifies contribution of x_j via

$$\phi_j(\mathbf{x}) = \frac{1}{|P|!} \sum_{\tau \in \Pi} \underbrace{\hat{f}_{S_j^\tau \cup \{j\}}(\mathbf{x}_{S_j^\tau \cup \{j\}}) - \hat{f}_{S_j^\tau}(\mathbf{x}_{S_j^\tau})}_{\Delta(j, S_j^\tau) \text{ marginal contribution of feature } j}$$



- **Interpretation:** $\phi_j(\mathbf{x})$ quantifies how much feature x_j contributes to the difference between $\hat{f}(\mathbf{x})$ and the mean prediction $\hat{f}_\emptyset$
 $\rightsquigarrow$ Marginal contributions and Shapley values can be negative
- **Exact computation of ϕ_j :** Using PD function $\hat{f}_S(\mathbf{x}_S) = \frac{1}{n} \sum_{i=1}^n \hat{f}(\mathbf{x}_S, \mathbf{x}_{-S}^{(i)})$ yields

$$\phi_j(\mathbf{x}) = \frac{1}{|P|!} \sum_{\tau \in \Pi} \frac{1}{n} \sum_{i=1}^n \hat{f}(\mathbf{x}_{S_j^\tau \cup \{j\}}, \mathbf{x}_{- \{S_j^\tau \cup \{j\}\}}^{(i)}) - \hat{f}(\mathbf{x}_{S_j^\tau}, \mathbf{x}_{-S_j^\tau}^{(i)})$$

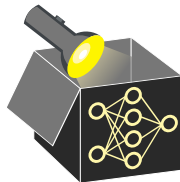
- $\rightsquigarrow \hat{f}_S$ marginalizes over all features not in S using all obs. $i = 1, \dots, n$
- $\rightsquigarrow$ Exact computation requires $|P|! \cdot n$ marginal contribution terms

ESTIMATION: A PRACTICAL PROBLEM

- **Exact computation is infeasible for many features:**

For $|P| = 10$, the number of permutations is $10! \approx 3.6$ million

↪ Complexity grows factorially with feature count



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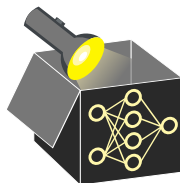
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Each permut. τ defines a coal. S_j^τ needing its own estimate of $\hat{f}_{S_j^\tau}(\mathbf{x}_{S_j^\tau})$

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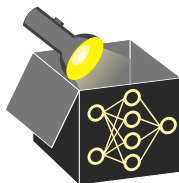
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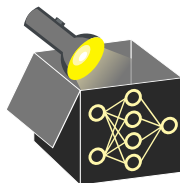
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- **Tradeoff: Accuracy vs. Efficiency**

Larger M improves Shapley approximation

↪ Higher cost, but better fidelity to the exact value

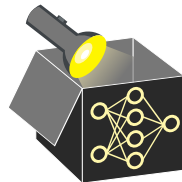


APPROXIMATION ALGORITHM

► “Strumbelj et al.” 2014

Estimate Shapley value ϕ_j of observation $\mathbf{x}$ for feature j :

- **Input:** $\mathbf{x}$ obs. of interest, j feat. of interest, $\hat{f}$ model, $\mathcal{D}$ data, M iterations

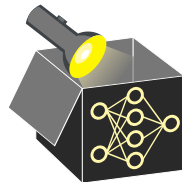


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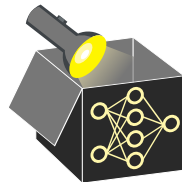


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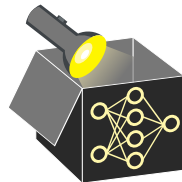


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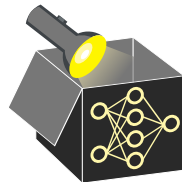


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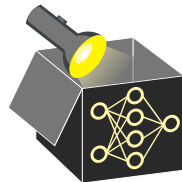


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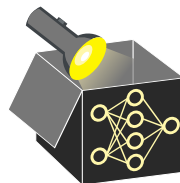


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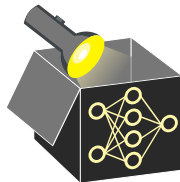
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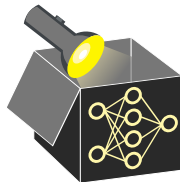
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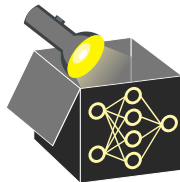
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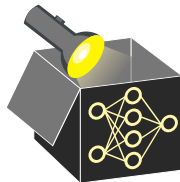
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- ⑤ Compute marginal contribution $\Delta(j, S_m) = \hat{f}(\mathbf{x}_{+j}^{(m)}) - \hat{f}(\mathbf{x}_{-j}^{(m)})$





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 - ② Compute Shapley value $\phi_j = \frac{1}{M} \sum_{m=1}^M \Delta(j, S_m)$
- $\rightsquigarrow$ Over M iterations, the PD functions $\hat{f}_{S_m}(\mathbf{x}_{S_m})$ and $\hat{f}_{S_m \cup \{j\}}(\mathbf{x}_{S_m \cup \{j\}})$ are approximated by $\hat{f}(\mathbf{x}_{-j}^{(m)})$ and $\hat{f}(\mathbf{x}_{+j}^{(m)})$, where features not in the coalition (to be marginalized) are imputed with vals from random data points $\mathbf{z}^{(m)}$

SHAPLEY VALUE APPROX. - ILLUSTRATION

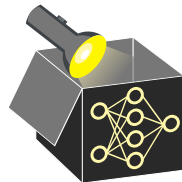
Definition

$\mathbf{x}$: obs. of interest

$\mathbf{x}$ with feature values in $\mathbf{x}_{S_m}$ (other are replaced)

$$\phi_j(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^M [\hat{f}(\mathbf{x}_{+j}^{(m)}) - \hat{f}(\mathbf{x}_{-j}^{(m)})]$$

$\mathbf{x}$ with feature values in $\mathbf{x}_{S_m \cup \{j\}}$



	Temperature	Humidity	Windspeed	Year
$\mathbf{x}$	10.66	56	11	2012
$\mathbf{x}_{+j}$	10.66	56	random : $z_{windspeed}^{(m)}$	2012
$\mathbf{x}_{-j}$	10.66	56	random : $z_{windspeed}^{(m)}$	random : $z_{year}^{(m)}$
			$\underbrace{\hspace{10em}}_j$	

SHAPLEY VALUE APPROX. - ILLUSTRATION

Definition

Contribution of feature j
to coalition S_m

$$\phi_j(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^M \underbrace{\left[\hat{f}(\mathbf{x}_{+j}^{(m)}) - \hat{f}(\mathbf{x}_{-j}^{(m)}) \right]}_{:= \Delta(j, S_m)}$$

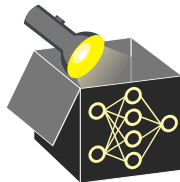
- $\Delta(j, S_m) = \hat{f}(\mathbf{x}_{+j}^{(m)}) - \hat{f}(\mathbf{x}_{-j}^{(m)})$ is marginal contribution of feature j to coalition S_m
- Here: Feature *year* contributes +700 bike rentals if it joins coalition $S_m = \{\text{temp}, \text{hum}\}$

	Temperature	Humidity	Windspeed	Year	Count
$\mathbf{x}$	10.66	56	11	2012	
$\mathbf{x}_{+j}$	10.66	56	random : $z_{\text{windspeed}}^{(m)}$	2012	5600
$\mathbf{x}_{-j}$	10.66	56	random : $z_{\text{windspeed}}^{(m)}$	random : $z_{\text{year}}^{(m)}$	4900

j

$\hat{f}$

$\Delta(j, S_m)$
marginal contribution

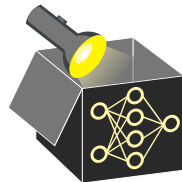


SHAPLEY VALUE APPROX. - ILLUSTRATION

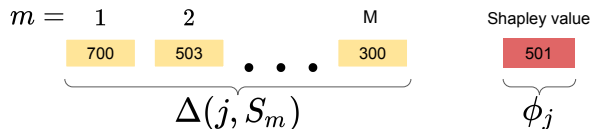
Definition

average the contributions
of feature j

$$\phi_j(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^M [\hat{f}(\mathbf{x}_{+j}^{(m)}) - \hat{f}(\mathbf{x}_{-j}^{(m)})]$$



- Compute marginal contribution of feature j towards the prediction across all randomly drawn feature coalitions $S_1, \dots, S_m$
- Average all M marginal contributions of feature j
- Shapley value ϕ_j is the payout of feature j , i.e., how much feature *year* contributed to the overall prediction in bicycle counts of a specific obs. $\mathbf{x}$



REVISITED: AXIOMS FOR FAIR ATTRIBUTIONS

We adapt the classic Shapley axioms to the setting of model predictions:

- **Efficiency:** Sum of Shapley values adds up to the centered prediction:

$$\sum_{j=1}^p \phi_j(\mathbf{x}) = \hat{f}(\mathbf{x}) - \mathbb{E}_{\mathbf{x}}[\hat{f}(\mathbf{x})]$$

↪ All predictive contribution is fully distributed among features

- **Symmetry:** Identical contributors receive equal value:

$$\hat{f}_{S \cup \{j\}}(\mathbf{x}_{S \cup \{j\}}) = \hat{f}_{S \cup \{k\}}(\mathbf{x}_{S \cup \{k\}}) \quad \forall S \subseteq P \setminus \{j, k\} \Rightarrow \phi_j = \phi_k$$

↪ Interaction effects are shared equitably

- **Dummy (Null Player):** Irrelevant features receive zero attribution:

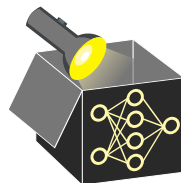
$$\hat{f}_{S \cup \{j\}}(\mathbf{x}_{S \cup \{j\}}) = \hat{f}_S(\mathbf{x}_S) \quad \forall S \subseteq P \Rightarrow \phi_j = 0$$

↪ Shapley value is zero for unused features (e.g., trees or LASSO)

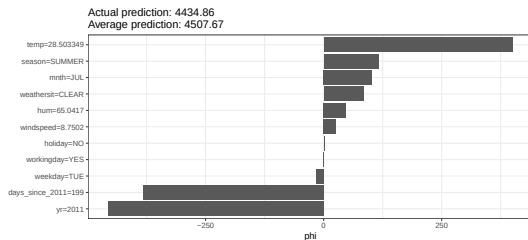
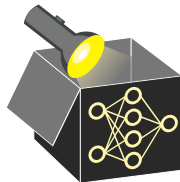
- **Additivity:** Attributions are additive across models:

$$\phi_j(v_1 + v_2) = \phi_j(v_1) + \phi_j(v_2)$$

↪ Enables combining Shapley values for model ensembles



BIKE SHARING DATASET



- Shapley decomposition for a single prediction in bike sharing dataset
- Model pred.: $\hat{f}(\mathbf{x}^{(200)}) = 4434.86$ vs. dataset avg.: $\mathbb{E}_{\mathbf{x}}[\hat{f}(\mathbf{x})] = 4507.67$
- Total feature attribution: $\sum_j \phi_j = -72.81$
 $\rightsquigarrow$ Explain downward shift from mean prediction
- Temperature (with value 28.5°C) – strongest positive contributor: $+400$
- $\text{yr} = 2011$ and $\text{days_since_2011} = 199$ strongly reduce prediction
 $\rightsquigarrow$ Model captures lower bike demand in 2011 compared to 2012

ADVANTAGES AND DISADVANTAGES

Advantages:

- **Strong theoretical foundation** from cooperative game theory
- **Fair attribution:** Prediction is additively distributed across features
~> Easy to interpret for users
- **Contrastive explanations:** Quantify each feature's role in deviating from the average prediction

Disadvantages:

- **Comput. cost:** Exact computation scales factorially with feature count
~> Without sampling, all 2^p coalitions (or $p!$ permuts) must be evaluated
- **Issue with correlated features:** Shapley values may evaluate the model on feature combinations that do not occur in the real data

